

Manan Mrig

mmrig.me | (612) 419-2000 | mmrig4@gmail.com | linkedin.com/in/manan-mrig

ML Engineer focused on building intelligent, scalable applications and the distributed systems that power them.

EDUCATION

University of Waterloo

Waterloo, ON

Master's of Data Science and Artificial Intelligence

Sep 2026 - Dec 2027

- Notable Coursework: Statistical Learning - Classification, Data Engineering, Machine Learning

University of Minnesota, Twin Cities

Minneapolis, MN

Bachelor's of Science, Computer Science

Sep 2020 - May 2024

- Notable Coursework: Machine Learning, Algorithms & Data Structures, Distributed Systems, Operating Systems

WORK EXPERIENCE

Software Engineer (Machine Learning Infrastructure)

June 2024 - July 2025

Target Corporation - Demand Forecasting Engine, Inventory Insights Acquisition

Minneapolis, MN

- Maintained and enhanced production GAMM-based forecasting pipelines for 100K+ SKUs using multi-year retail time-series data on Hadoop, supporting item- and chain-level inventory forecasting.
- Developed batch jobs and Kotlin Micronaut and Springboot microservices to enable human overrides of production forecasting models, giving planners greater control over inventory planning and physical inventory workflows.
- Owned production microservices supporting data-driven inventory interventions across 1,900+ stores, improving inventory reliability and system accuracy by 20% through ongoing enhancements and maintenance.

Teaching Assistant

January 2022 – May 2024

University of Minnesota, Twin Cities

Minneapolis, MN

- Led lab sections and office hours for Machine Architecture, Software Design for Data Scientists, Programming Concepts, and Discrete Mathematics, reinforcing concepts in operating systems, computer architecture, and systems programming.
- Designed, validated, published, and graded programming assignments in C, Python, Java, and Assembly while mentoring students through debugging and core computer science concepts.

Software Engineer Intern

June 2023 – August 2023

Target Corporation - Paystack

Minneapolis, MN

- Built Kotlin microservices for high-volume real-time payment processing, supporting low-latency analytics, monitoring, and reliable production payment workflows.
- Improved system performance by 40% through comprehensive unit testing and performance optimization, increasing throughput under production workloads.

Software Engineer Intern

May 2021 – July 2021

Centrl Inc.

Mountain View, CA

- Migrated backend components of a legacy PHP customer administration platform to Java and Spring Boot, reducing request processing time by 30%.
- Implemented OAuth2 authentication using Google's Authorization Framework in a Spring Boot application, streamlining secure access for 50+ internal users.

TECHNICAL SKILLS

Languages: Python, Kotlin, Java, SQL, C++, JavaScript, TypeScript, C, Bash

Machine Learning: PyTorch, TensorFlow, NumPy, Pandas, Statistical Modeling, Matplotlib

Backend: Spring Boot, Micronaut, Flask, LangChain, Java RMI

Infrastructure: Docker, Kafka, PostgreSQL, Hadoop, Hive, AWS, GCP, Git, Vercel, Jira

PROJECTS

Personal AI Assistant Ecosystem | *SwiftUI, Kotlin, Python, RunPod, LangChain, & Postgres*

2025

- Developed a multimodal AI assistant using SwiftUI, Kotlin, Python, LangChain, and RunPod-hosted LLMs, implementing long-term vector memory and reducing inference latency by 40%.

Pro-Posterous: Human-AI Co-Creation Canvas | *Next.js, tldraw, LangChain, Diffusion Models*

2025

- Built an AI-powered poster design platform integrating diffusion models (FLUX, WAN, SDXL) with an interactive canvas for real-time iterative image editing.